torch.amin#

https://pytorch.org/docs/stable/generated/torch.amin.html#torch.amin

Description:

Returns the minimum value of each slice of the input tensor in the given dimension(s) dim. amax/amin supports reducing on multiple dimensions, amax/amin does not return indices. Both amax/amin evenly distribute gradients between equal values when there are multiple input elements with the same minimum or maximum value.

Function Signature

```
torch.amin(input, dim, keepdim=False, *, out=None) → Tensor#
The difference between max/min and amax/amin is:
For max/min:
Parameters
Keyword Arguments
```

Parameters

Keyword

out (Tensor, optional) – the output tensor.

Code Examples

```
>>> a = torch.randn(4, 4)
>>> a
tensor([[ 0.6451, -0.4866,  0.2987, -1.3312],
        [-0.5744,  1.2980,  1.8397, -0.2713],
        [ 0.9128,  0.9214, -1.7268, -0.2995],
        [ 0.9023,  0.4853,  0.9075, -1.6165]])
>>> torch.amin(a, 1)
tensor([-1.3312, -0.5744, -1.7268, -1.6165])
```

Notes & Warnings

NOTE

Note The difference between max/min and amax/amin is: amax/amin supports reducing on multiple dimensions, amax/amin does not return indices. Both amax/amin evenly distribute gradients between equal values when there are multiple input elements with the same minimum or maximum value. For max/min: If reduce over all dimensions(no dim specified), gradients evenly distribute between equally max/min values. If reduce over one specified axis, only propagate to the indexed element.

Detailed Documentation

Documentation

Returns the minimum value of each slice of the input tensor in the given dimension(s) dim.

amax/amin supports reducing on multiple dimensions,

amax/amin does not return indices.

Both amax/amin evenly distribute gradients between equal values when there are multiple input elements with the same minimum or maximum value.

If reduce over all dimensions(no dim specified), gradients evenly distribute between equally max/min values.

If reduce over one specified axis, only propagate to the indexed element.

If keepdim is True, the output tensor is of the same size as input except in the dimension(s) dim where it is of size 1. Otherwise, dim is squeezed (see torch.squeeze()), resulting in the output tensor having 1 (or len(dim)) fewer dimension(s).

input (Tensor) – the input tensor.

dim (int or tuple of ints, optional) – the dimension or dimensions to reduce. If None, all dimensions are reduced.

keepdim (bool, optional) – whether the output tensor has dim retained or not. Default: False.

out (Tensor, optional) – the output tensor.
